

TwinGuard Aero

From an engine reading to a reviewable decision.

An explainable digital-twin demonstrator linking measurement trust, engine condition and mission-profile consequences.

01

Trust the evidence

Freshness, quality and related-signal consistency.

02

Explain the condition

Healthy reference, residuals and temporal evidence.

03

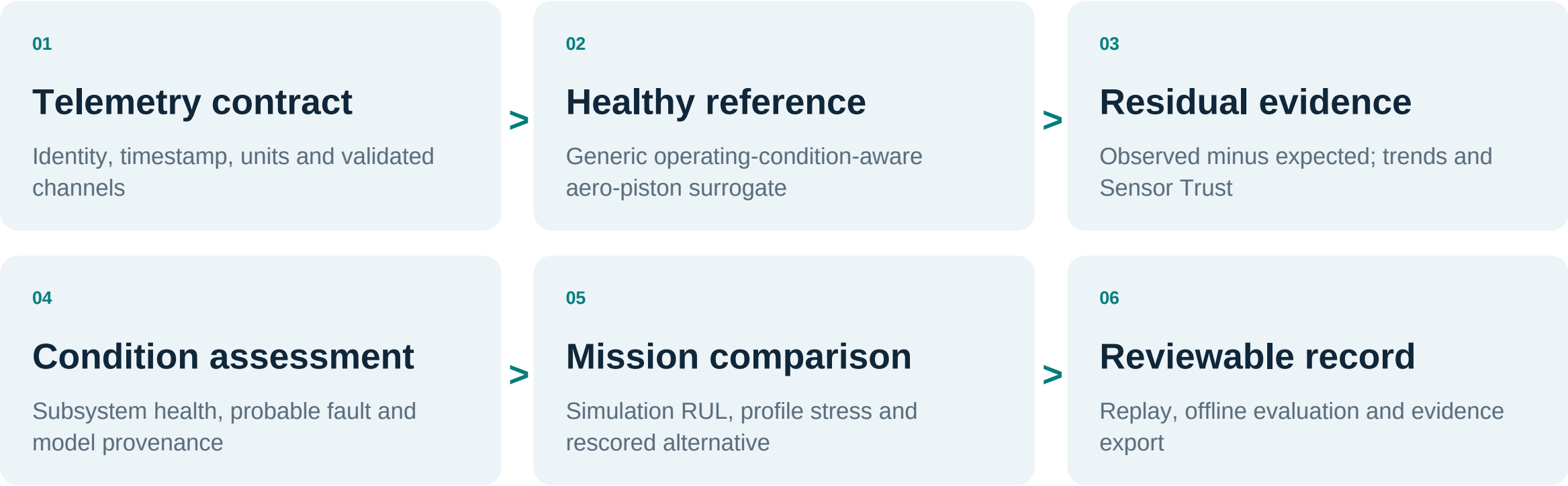
Compare the profile

Same mission, changed engine state, rescored alternative.

CURRENT SCOPE Synthetic proof-of-concept. Generic engine model. Human engineering review.

The analytical twin is behind the interface.

One canonical state connects data integrity, expected behavior, diagnosis and engineering review.



Data gate: stale or low-quality live state becomes DATA HOLD. A connected browser alone is not evidence of fresh measurements.

A repeatable scenario, with a record of what changed.

Hosted Evaluation Center: baseline, progressive fault, comparison and exported evidence.

1

Capture baseline

Start from the healthy browser simulator and preserve its complete state.

2

Apply lubrication loss

Choose the fault and intensity. Observe the progressive synthetic change.

3

Inspect + capture

Compare pressure residual, vibration, health and RUL with baseline.

4

Export evidence

Download state, units, provenance, scenario inputs and a SHA-256 checksum.

What this evidence can establish

The recorded state and model assumptions behind this demonstrator run. It supports inspection and repeatability.

What it cannot establish

Independent classifier accuracy or real-engine validity. A checksum detects changes; it does not authenticate the data source.

Make the model assumptions inspectable.

A confident presentation distinguishes implementation evidence from operational validation.

Healthy reference	Generic low-order surrogate	Calibrate with authorized maps and independent rig data
Diagnostic runtime	Engineering fallback or compatible synthetic ML	Show the active mode; do not imply an unused model ran
Remaining useful life	Simulation-derived estimate and sensitivity band	Validate against longitudinal degradation / maintenance evidence
Mission feasibility	Engineering index for profile comparison	Not a calibrated probability of mission success
CAN integration	Configurable adapter and canonical schema	Needs authorized DBC, channel validation and rig testing

The simulator and healthy-reference model share assumptions. Synthetic agreement alone does not establish real-engine generalization.

The next strongest proof is an independent recording.

Compare methods on the same held-out runs, preserve failure cases and report the denominator.

EXPERIMENT DESIGN

Threshold baseline vs hybrid twin

Compare fixed thresholds, operating-context residuals, temporal evidence and the complete Sensor Trust workflow.

LEAKAGE CONTROL

Hold out complete runs

Separate operating conditions and ultimately engines. Do not randomly split adjacent samples from the same fault trajectory.

False alarms	Per operating hour, with normal transients included
Fault detection	Per-fault precision / recall and missed detections
Useful warning	Time to stable alert and lead time to labeled event
RUL / uncertainty	Error and coverage only where reference truth exists
Deployment cost	p50/p95 latency, CPU and memory at measured load

Acceptance targets should be agreed with the target-engine partner before final testing. No unmeasured performance target is presented as an achieved result.

Accessible today. Engine-specific validation next.

Two explicit execution modes keep the presentation accessible and the technical boundary honest.

HOSTED DEMONSTRATION

Browser synthetic runtime

An accessible evaluator workflow with controlled scenarios. It is not a live ECU connection.

LOCAL / CONTROLLED ENVIRONMENT

Backend integration workflow

FastAPI, authenticated APIs, WebSockets and persistence. Current scope: one configured engine and process.

NOW

Reproducible release

Source, working flows, versioned software checks and synthetic evidence.

NEXT

Authorized read-only rig

Confirm channels and timing; calibrate healthy behavior; collect independent labels.

THEN

Validated deployment

Held-out diagnosis and RUL studies, HIL, recovery/security tests and domain assurance.

PS identity and high-level scope: sih.gov.in/sih2026PS

A separate draft follows the verified SIH2026 template. Its team ID must be confirmed before portal submission.